TidyTuesday Week 42

Pranay Gundam

Friday 16th October, 2026

Contents

1 Weekly Summary **2**

2 Date: 2026-10-12 **3**
2.1 Raw Regression 3
2.2 Detrended Regression 5

3 Date: 2026-10-13 **7**
3.1 Raw Regression 7
3.2 Detrended Regression 9

4 Date: 2026-10-14 **11**
4.1 Raw Regression 11
4.2 Detrended Regression 13

5 Date: 2026-10-15 **15**
5.1 Raw Regression 15
5.2 Detrended Regression 17

6 Date: 2026-10-16 **19**
6.1 Raw Regression 19
6.2 Detrended Regression 21

1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to the weekly ritual where we throw two unrelated economic series into a blender and solemnly report what sprays out. This week the dice gave us a genuinely mixed bag, which is to say: mostly noise, one absolute showstopper, and a couple of trend-mirages dressed up to look important. Grab a beverage.

Monday kicked us off with Belize's exchange rate versus the (discontinued, naturally) U.S. Balance on Services. The raw regression squeaked past the significance line with an R-squared of about nine percent, which is the statistical equivalent of a participation trophy. But here's the twist: once we detrended both series, the relationship didn't collapse – it actually got slightly stronger and flipped sign. So Belizean currency and American services have a faint, stubborn little wiggle-for-wiggle thing going on that survives trend removal. Do I have the faintest idea why? Absolutely not. Am I going to pretend it means something? Only ironically.

Then Tuesday happened, and Tuesday was glorious. Gasoline prices in the South Census Region (Size Class B, because precision matters) versus 30-year FHA mortgage rates. The raw fit was decent, but the detrended version came back with an R-squared of 0.63 and a p-value with fifty-one zeros after the decimal point. That is, by the standards of this deeply unserious project, a tidal wave. A relationship this strong surviving detrending is actually the rare "huh, maybe this is real" case – gas prices and mortgage rates both dance to the tune of inflation and the broader cost of money, so a genuine link here is at least theorizing-eligible rather than pure cosmic coincidence. I'll allow myself one (1) raised eyebrow.

The back half of the week brought us crashing back to earth, as is tradition. Wednesday paired corporate machinery cash deposits with government-sponsored enterprise debt, and the raw regression screamed an 80 percent R-squared – which, the instant we detrended, evaporated into a limp, insignificant nothing. Textbook trend mirage: two numbers that both go up over time holding hands and calling it love. Thursday's Swedish government consumption prices versus Kansas City energy services told the same sad story (strong raw, dead on detrending), and Friday's corporate bond index against Middle Atlantic unemployment couldn't even muster a convincing fib – the detrended p-value was a sleepy 0.79. As always, the standard disclaimer applies in bold invisible ink: these are randomly paired series, nothing here is causal, and spurious correlation isn't the exception, it's the entire business model. See you next week for more.

2 Date: 2026-10-12

Series ID: BOPBSVN
This series is titled Balance on Services (DISCONTINUED) and has a frequency of Quarterly. The units are Billions of Dollars and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1960-01-01 and the observation end date is 2014-01-01.The popularity of this series is 1.

Series ID: XRNCUSBZA618NRUG
This series is titled Exchange Rate (market+estimated) for Belize and has a frequency of Annual. The units are National Currency Units per US Dollar and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1970-01-01 and the observation end date is 2023-01-01.The popularity of this series is 1.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_XRNCUSBZA618NRUG	**R-squared:**	0.090
Model:	OLS	**Adj. R-squared:**	0.069
Method:	Least Squares	**F-statistic:**	4.246
Date:	Mon, 12 Oct 2026	**Prob (F-statistic):**	0.0454
Time:	20:28:58	**Log-Likelihood:**	34.026
No. Observations:	45	**AIC:**	-64.05
Df Residuals:	43	**BIC:**	-60.44
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	1.9274	0.024	79.937	0.000	1.879	1.976
value_fred_BOPBSVN	0.0023	0.001	2.060	0.045	4.83e-05	0.004

Omnibus:	17.591	**Durbin-Watson:**	0.425
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	23.370
Skew:	-1.275	**Prob(JB):**	8.42e-06
Kurtosis:	5.441	**Cond. No.**	30.5

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

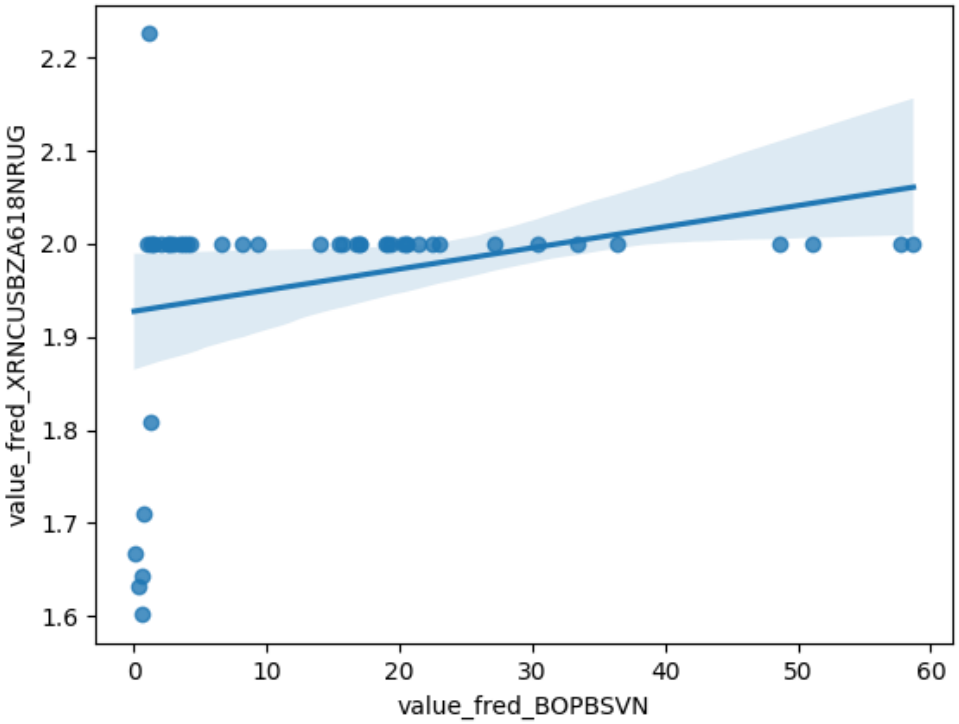


Figure 1: Raw Regression Plot for 2026-10-12

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_XRNCUSBZA618NRUG_detrended	**R-squared:**	0.123
Model:	OLS	**Adj. R-squared:**	0.103
Method:	Least Squares	**F-statistic:**	6.041
Date:	Mon, 12 Oct 2026	**Prob (F-statistic):**	0.0181
Time:	20:28:58	**Log-Likelihood:**	41.263
No. Observations:	45	**AIC:**	-78.53
Df Residuals:	43	**BIC:**	-74.91
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	5.671e-15	0.015	3.84e-13	1.000	-0.030	0.030
value_fred_BOPBSVN_detrended	-0.0050	0.002	-2.458	0.018	-0.009	-0.001

Omnibus:	8.915	**Durbin-Watson:**	0.588
Prob(Omnibus):	0.012	**Jarque-Bera (JB):**	16.952
Skew:	0.296	**Prob(JB):**	0.000208
Kurtosis:	5.948	**Cond. No.**	7.28

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

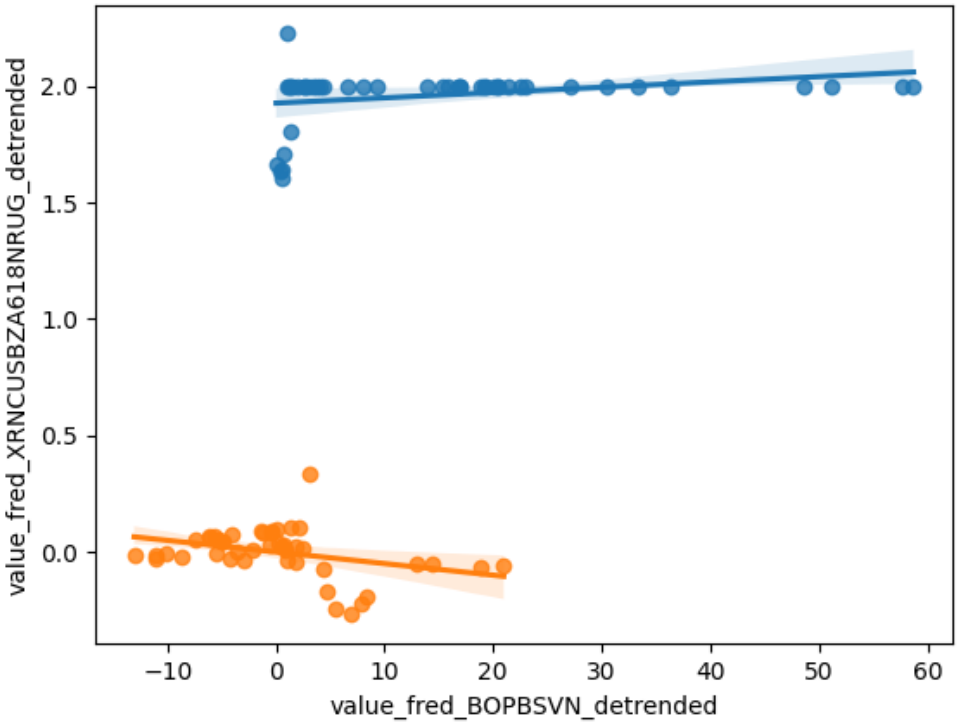


Figure 2: Detrended Regression Plot for 2026-10-12

3 Date: 2026-10-13

Series ID: APUB30074714
This series is titled Average Price: Gasoline, Unleaded Regular (Cost per Gallon/3.785 Liters) in the South Census Region - Size Class B and has a frequency of Monthly. The units are U.S. Dollars and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1978-01-01 and the observation end date is 1997-12-01.The popularity of this series is 1.

Series ID: FHA30
This series is titled 30-Year FHA Mortgage Rate: Secondary Market (DISCONTINUED) and has a frequency of Monthly. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1964-01-01 and the observation end date is 2000-06-01.The popularity of this series is 9.

3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_FHA30	**R-squared:**	0.169
Model:	OLS	**Adj. R-squared:**	0.165
Method:	Least Squares	**F-statistic:**	47.29
Date:	Tue, 13 Oct 2026	**Prob (F-statistic):**	5.59e-11
Time:	18:35:32	**Log-Likelihood:**	-534.79
No. Observations:	235	**AIC:**	1074.
Df Residuals:	233	**BIC:**	1081.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	3.5713	1.048	3.408	0.001	1.507	5.636
value_fred_APUB30074714	6.5135	0.947	6.876	0.000	4.647	8.380

Omnibus:	11.293	**Durbin-Watson:**	0.039
Prob(Omnibus):	0.004	**Jarque-Bera (JB):**	6.092
Skew:	0.195	**Prob(JB):**	0.0476
Kurtosis:	2.315	**Cond. No.**	13.6

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

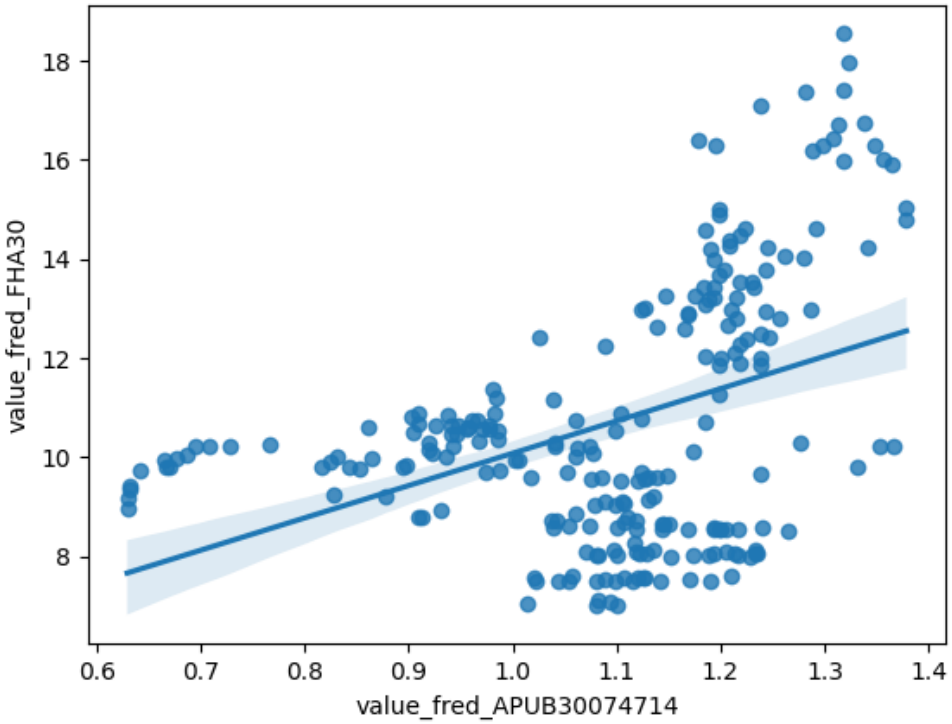


Figure 3: Raw Regression Plot for 2026-10-13

3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_FHA30_detrended	**R-squared:**	0.626
Model:	OLS	**Adj. R-squared:**	0.624
Method:	Least Squares	**F-statistic:**	389.6
Date:	Tue, 13 Oct 2026	**Prob (F-statistic):**	1.23e-51
Time:	18:35:32	**Log-Likelihood:**	-352.60
No. Observations:	235	**AIC:**	709.2
Df Residuals:	233	**BIC:**	716.1
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	5.561e-13	0.071	7.82e-12	1.000	-0.140	0.140
value_fred_APUB30074714_detrended	8.7393	0.443	19.738	0.000	7.867	9.612

Omnibus:	5.577	**Durbin-Watson:**	0.209
Prob(Omnibus):	0.062	**Jarque-Bera (JB):**	5.236
Skew:	0.334	**Prob(JB):**	0.0729
Kurtosis:	3.299	**Cond. No.**	6.23

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

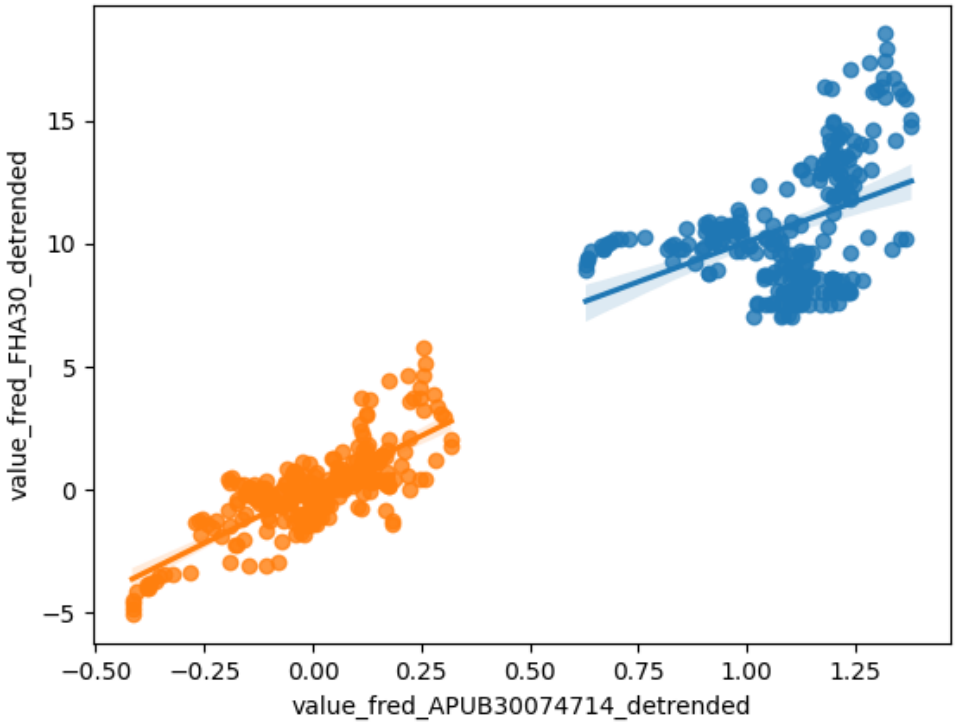


Figure 4: Detrended Regression Plot for 2026-10-13

4 Date: 2026-10-14

Series ID: QFR201333USNO
This series is titled Quarterly Financial Report: U.S. Corporations: Machinery: Cash and Demand Deposits in the U.S. and has a frequency of Quarterly. The units are Millions of Dollars and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 2000-10-01 and the observation end date is 2026-04-01.The popularity of this series is 1.

Series ID: GSETCMDODFS
This series is titled Government-Sponsored Enterprises; Debt Securities and Loans; Liability, Level and has a frequency of Quarterly, End of Period. The units are Millions of U.S. Dollars and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1945-10-01 and the observation end date is 2026-04-01.The popularity of this series is 2.

4.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_GSETCMDODFS	**R-squared:**	0.805
Model:	OLS	**Adj. R-squared:**	0.803
Method:	Least Squares	**F-statistic:**	416.8
Date:	Wed, 14 Oct 2026	**Prob (F-statistic):**	1.26e-37
Time:	16:02:51	**Log-Likelihood:**	-1578.2
No. Observations:	103	**AIC:**	3160.
Df Residuals:	101	**BIC:**	3166.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	4.816e+05	2.74e+05	1.758	0.082	-6.18e+04	1.02e+06
value_fred_QFR201333USNO	237.3381	11.626	20.415	0.000	214.276	260.400

Omnibus:	4.636	**Durbin-Watson:**	0.398
Prob(Omnibus):	0.098	**Jarque-Bera (JB):**	4.489
Skew:	-0.511	**Prob(JB):**	0.106
Kurtosis:	2.949	**Cond. No.**	5.94e+04

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 5.94e+04. This might indicate that there are
strong multicollinearity or other numerical problems.

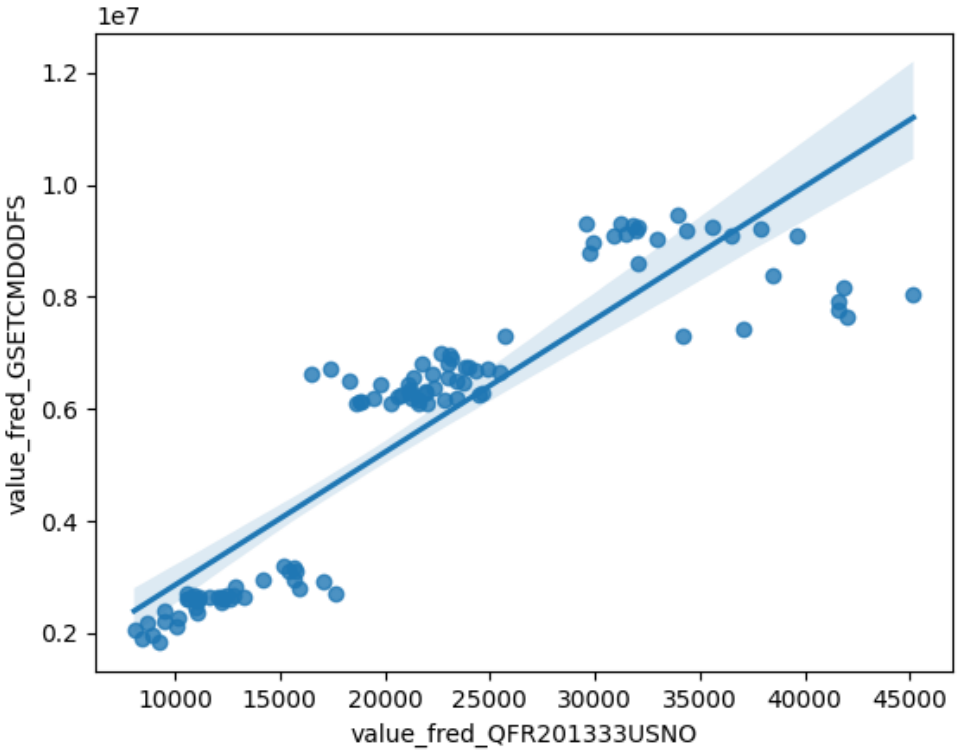


Figure 5: Raw Regression Plot for 2026-10-14

4.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_GSETCMDODFS_detrended	**R-squared:**	0.022
Model:	OLS	**Adj. R-squared:**	0.013
Method:	Least Squares	**F-statistic:**	2.317
Date:	Wed, 14 Oct 2026	**Prob (F-statistic):**	0.131
Time:	16:02:51	**Log-Likelihood:**	-1530.9
No. Observations:	103	**AIC:**	3066.
Df Residuals:	101	**BIC:**	3071.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-4.732e-09	6.86e+04	-6.9e-14	1.000	-1.36e+05	1.36e+05
value_fred_QFR201333USNO_detrended	28.1415	18.489	1.522	0.131	-8.536	64.819

Omnibus:	13.571	**Durbin-Watson:**	0.344
Prob(Omnibus):	0.001	**Jarque-Bera (JB):**	16.691
Skew:	0.705	**Prob(JB):**	0.000237
Kurtosis:	4.378	**Cond. No.**	3.71e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 3.71e+03. This might indicate that there are strong multicollinearity or other numerical problems.

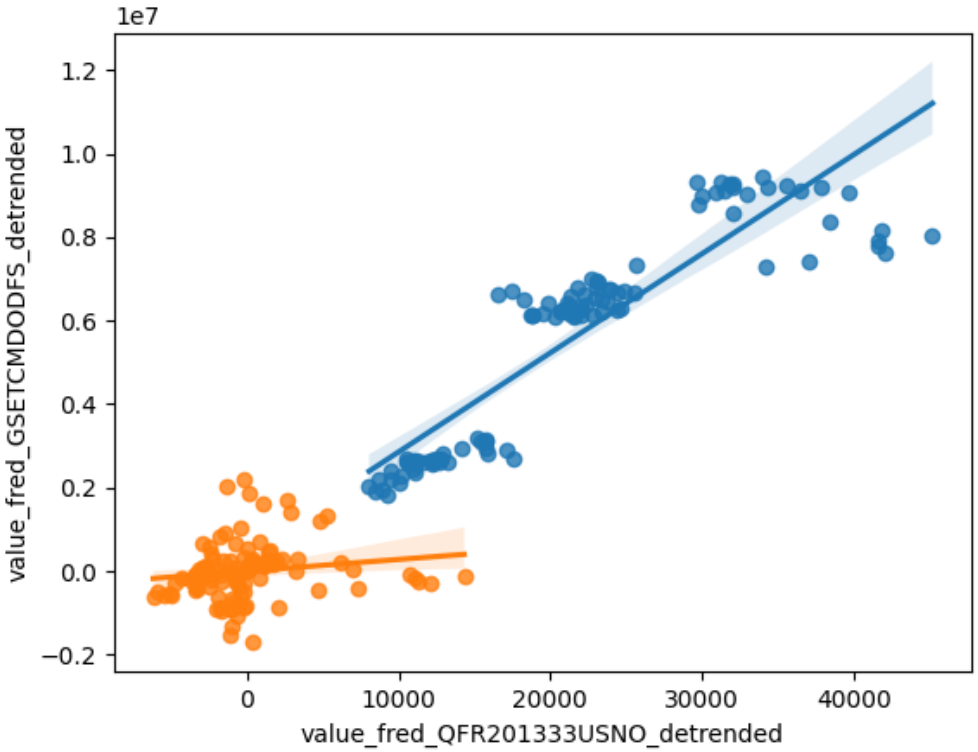


Figure 6: Detrended Regression Plot for 2026-10-14

5 Date: 2026-10-15

Series ID: PLGCPPSEA670NRUG
This series is titled Price Level of Government Consumption for Sweden and has a frequency of Annual. The units are Price Level of USA Output-side GDP in 2021=1 and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1950-01-01 and the observation end date is 2023-01-01.The popularity of this series is 1.

Series ID: CUUSA214SEHF
This series is titled Consumer Price Index for All Urban Consumers: Energy Services in Kansas City, MO-KS (CMSA) (DISCONTINUED) and has a frequency of Semiannual. The units are Index 1982-1984=100 and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1984-01-01 and the observation end date is 2017-07-01.The popularity of this series is 1.

5.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_CUUSA214SEHF	**R-squared:**	0.326
Model:	OLS	**Adj. R-squared:**	0.305
Method:	Least Squares	**F-statistic:**	15.50
Date:	Thu, 15 Oct 2026	**Prob (F-statistic):**	0.000418
Time:	15:51:08	**Log-Likelihood:**	-162.19
No. Observations:	34	**AIC:**	328.4
Df Residuals:	32	**BIC:**	331.4
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	51.5366	22.372	2.304	0.028	5.966	97.107
value_fred_PLGCPPSEA670NRUG	84.4492	21.453	3.937	0.000	40.752	128.147

Omnibus:	0.731	**Durbin-Watson:**	0.224
Prob(Omnibus):	0.694	**Jarque-Bera (JB):**	0.812
Skew:	0.276	**Prob(JB):**	0.666
Kurtosis:	2.482	**Cond. No.**	8.76

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

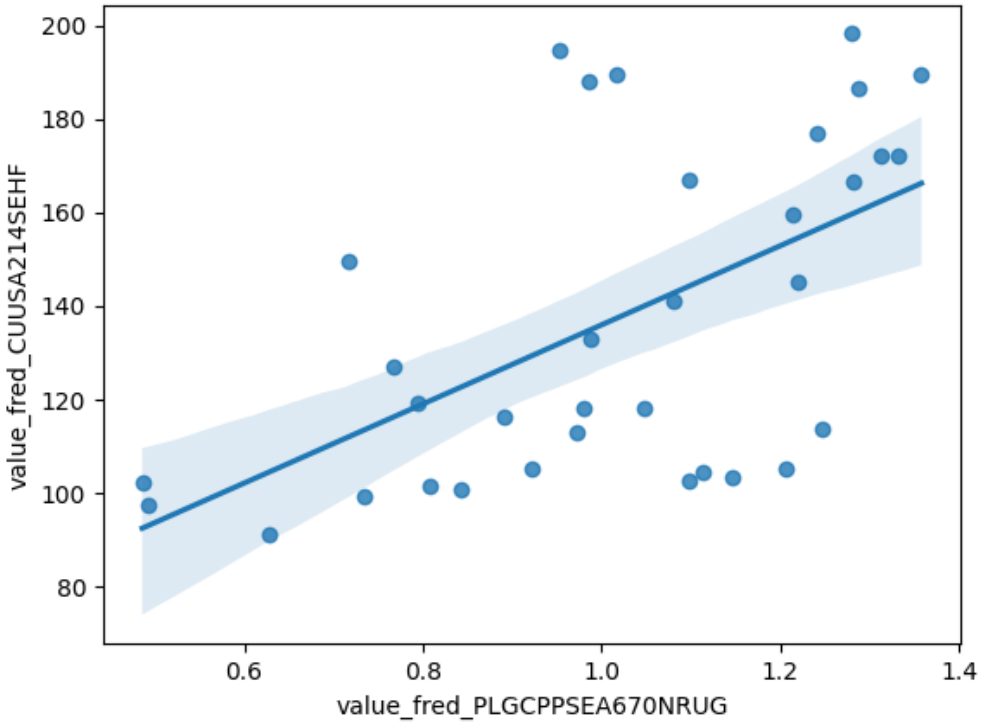


Figure 7: Raw Regression Plot for 2026-10-15

5.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_CUUSA214SEHF_detrended	**R-squared:**	0.006
Model:	OLS	**Adj. R-squared:**	-0.025
Method:	Least Squares	**F-statistic:**	0.1907
Date:	Thu, 15 Oct 2026	**Prob (F-statistic):**	0.665
Time:	15:51:08	**Log-Likelihood:**	-126.43
No. Observations:	34	**AIC:**	256.9
Df Residuals:	32	**BIC:**	259.9
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	1.842e-12	1.763	1.05e-12	1.000	-3.590	3.590
value_fred_PLGCPPSEA670NRUG_detrended	-4.1498	9.502	-0.437	0.665	-23.505	15.205

Omnibus:	0.585	**Durbin-Watson:**	1.001
Prob(Omnibus):	0.746	**Jarque-Bera (JB):**	0.498
Skew:	0.275	**Prob(JB):**	0.780
Kurtosis:	2.781	**Cond. No.**	5.39

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

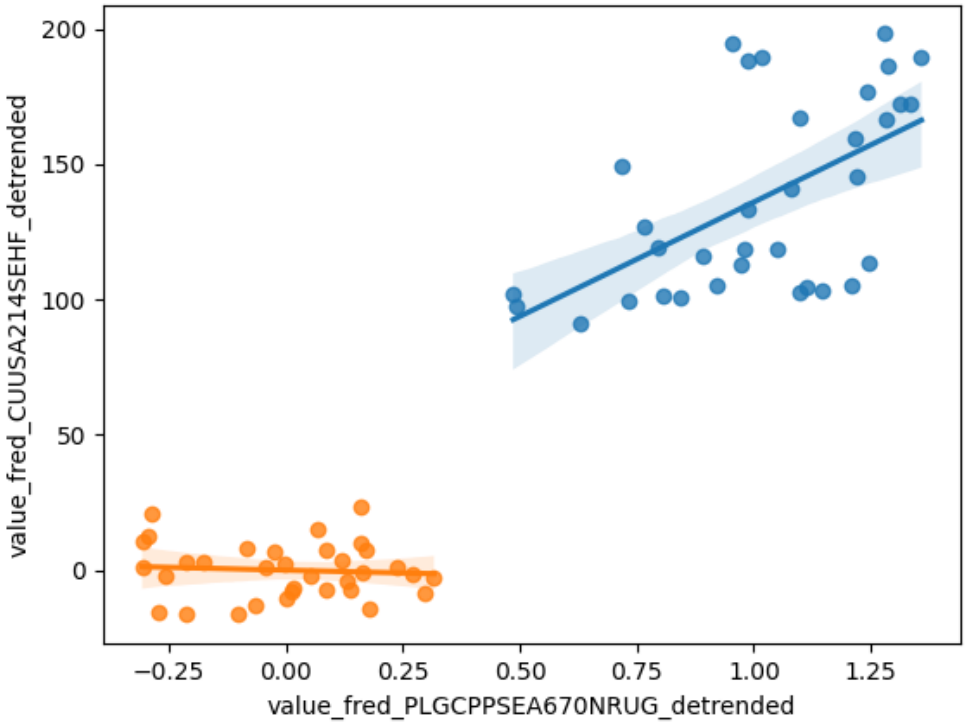


Figure 8: Detrended Regression Plot for 2026-10-15

6 Date: 2026-10-16

Series ID: BAMLCC8A015PYTRIV
This series is titled ICE BofA 15+ Year US Corporate Index Total Return Index Value and has a frequency of Daily, Close. The units are Index and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 2023-10-17 and the observation end date is 2026-10-15.The popularity of this series is 10.

Series ID: CMATUR
This series is titled Unemployment Rate in Middle Atlantic division and has a frequency of Monthly. The units are Percent and the seasonal adjustment is Seasonally Adjusted.The observation start date is 1976-01-01 and the observation end date is 2026-08-01.The popularity of this series is 2.

6.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_CMATUR	**R-squared:**	0.324
Model:	OLS	**Adj. R-squared:**	0.286
Method:	Least Squares	**F-statistic:**	8.618
Date:	Fri, 16 Oct 2026	**Prob (F-statistic):**	0.00884
Time:	15:39:36	**Log-Likelihood:**	5.6116
No. Observations:	20	**AIC:**	-7.223
Df Residuals:	18	**BIC:**	-5.232
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	1.8823	0.838	2.245	0.038	0.121	3.643
value_fred_BAMLCC8A015PYTRIV	0.0006	0.000	2.936	0.009	0.000	0.001

Omnibus:	1.277	**Durbin-Watson:**	0.475
Prob(Omnibus):	0.528	**Jarque-Bera (JB):**	1.121
Skew:	0.506	**Prob(JB):**	0.571
Kurtosis:	2.433	**Cond. No.**	8.54e+04

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 8.54e+04. This might indicate that there are
strong multicollinearity or other numerical problems.

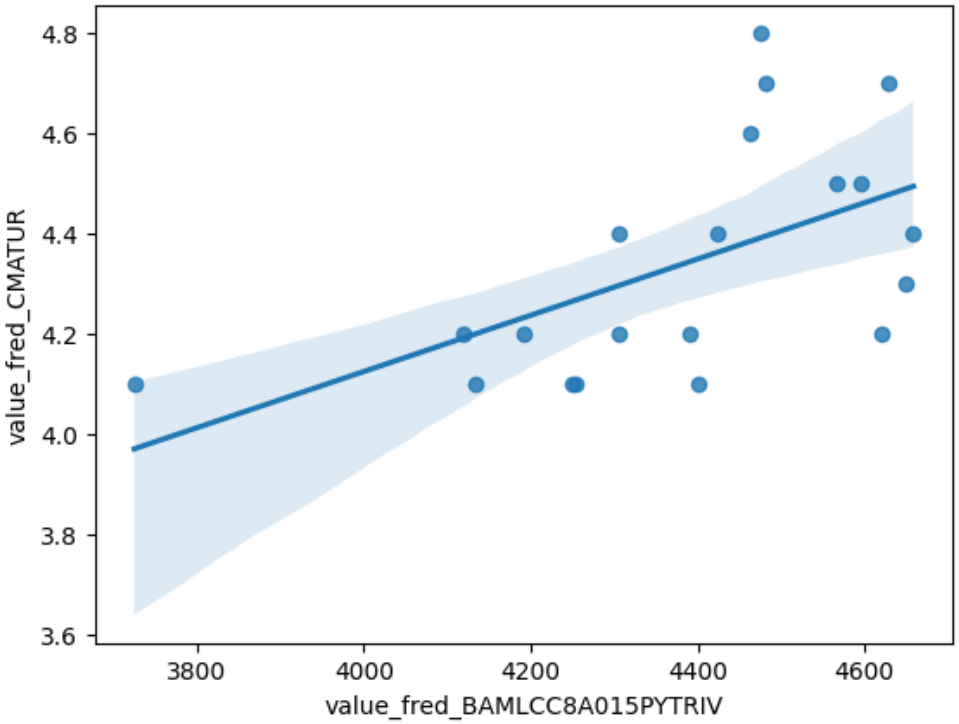


Figure 9: Raw Regression Plot for 2026-10-16

6.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_CMATUR_detrended	**R-squared:**	0.004
Model:	OLS	**Adj. R-squared:**	-0.051
Method:	Least Squares	**F-statistic:**	0.07620
Date:	Fri, 16 Oct 2026	**Prob (F-statistic):**	0.786
Time:	15:39:36	**Log-Likelihood:**	9.8365
No. Observations:	20	**AIC:**	-15.67
Df Residuals:	18	**BIC:**	-13.68
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-2.495e-13	0.035	-7.15e-12	1.000	-0.073	0.073
value_fred_BAMLCC8A015PYTRIV_detrended	-7.098e-05	0.000	-0.276	0.786	-0.001	0.000

Omnibus:	2.989	**Durbin-Watson:**	0.437
Prob(Omnibus):	0.224	**Jarque-Bera (JB):**	1.399
Skew:	0.594	**Prob(JB):**	0.497
Kurtosis:	3.517	**Cond. No.**	136.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

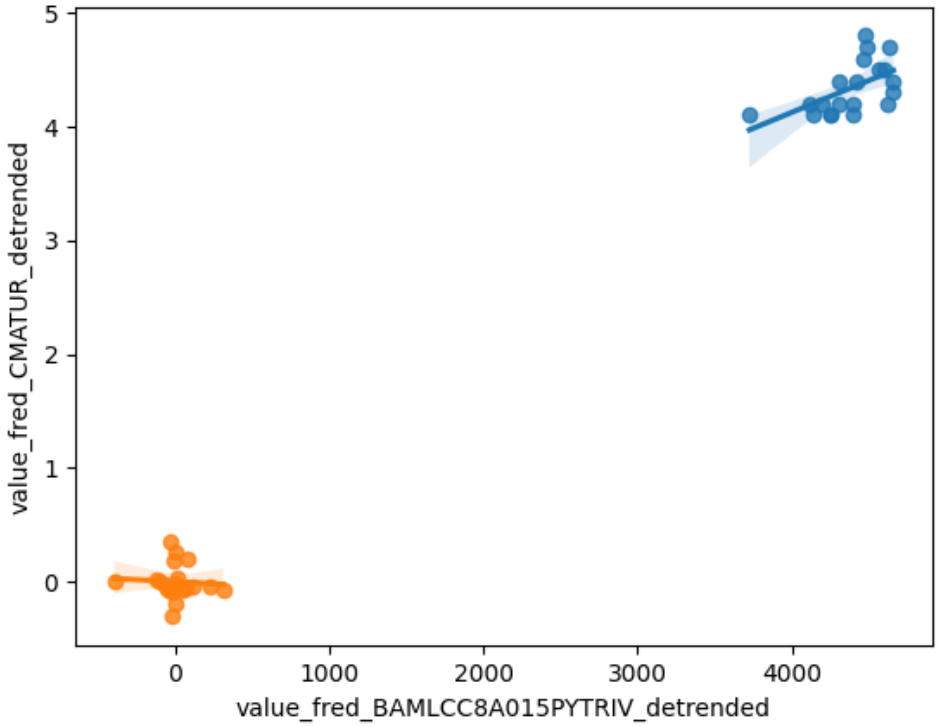


Figure 10: Detrended Regression Plot for 2026-10-16